MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6829

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, medium Mu, computer

Outline: 6-3 (EIA)
Base: E9-1
Envelope: T6-1/2
Cathode: Coated unipotential
Base connections:

Pin No.:	1	2	3	4	5	6	7	8	9
Element:	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk/k	Rg/g	Ik/k	Ic/g	Pp/a	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	C	ft
Maximum:	13.2	275	1, -50	100	---	0.1	21	1.0	2.3	140	Note 1
	6.6	---	---	---	---	Note 2	---	---	Note 3	---	---
Minimum	12.0	---	---	---	---	---	---	---	---	---	---
	6.0	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3 (Note 4)	150	0	0	220	---	---	---	---	---	---

Pulse service

Parameter:	Ef	Eb	Ec	eb	egk	ehk	Pg/g	Pp/a	ik/k	Du	tp	prr
Unit:	V	Vdc	Vdc	v	v	v	W	W	ma	%	us	---
Maximum:	13.2	275	---	550	10,-100	150	0.1	2.2	[			
	6.6	---	---	---	---	Note 5	---	---				
Minimum:	12.0	---	---	---	---	---	---	---		See figure 1		
	6.0	---	---	---	---	---	---	---	]			
TEST CONDITIONS:	6.3 (Note 4)	150	-20	---	10	---	---	---	---	1.0	10	1,000

GENERAL:

Qualification - Required

6829

FSC 5960

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	425	475	mA
1336	Heater-cathode leakage	6		0.4	II	Ihk	---	7	µAdc
1266	Total grid current	6,7	Rg = 0.5 Meg	0.4	II	Ic	0	-0.5	µAdc
1256	Electrode current (1) (anode)	6,8	Eb = 100 Vdc; Rk = 0	0.4	II	Ib	12.0	22.0	mAdc
1261	Electrode voltage (grid)	6	Ec/Ib = 150 µAdc; Rk = 0	0.4	II	Ec	---	-7.5	Vdc
1261	Electrode voltage (grid) (difference between sections)	6	Ec/Ib = 150 µAdc; Rk = 0	0.4	II	ΔEc	---	1.5	Vdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	6		2.5	I	R	200	---	Meg
1256	Electrode current (1) (anode)(change in individual tubes)	6,8	Ef = 5.7 V; Eb = 100 Vdc; Rk = 0	2.5	I	ΔIb Ef	---	15	%
1266	Grid emission	4,6,9	Ef = 7.5 V; Ec = -30 Vdc; Rg = 0.5 Meg	2.5	I	Ic	0	-1.0	µAdc
2201	Noise and microphonics	4,10,11	Ef = 6.3 V; Ebb = 250 Vdc; Eca[2] = 200 mVac; Rp = 0.1 Meg; Rk = 1,000	2.5	I	---	---	---	---
1256	Electrode current (2) (anode)	6		2.5	I	Ib	6.3	10.7	mAdc
1306	Transconductance	6	Ck = 1,000 µF	2.5	I	Sm	5,300	8,100	µmhos
1316	Amplification factor	6		---	---	Mu	39	55	---
---	Pulse cathode current	6,12	Rk = 1.0 ±0.01 ohm	2.5	Code H	Ik Δik (tp)	160 ---	--- 10	ma %
1331	Direct-interelectrode capacitance	6	No shield			Cga	2.4	3.6	pF
		6	No shield			Cin	3.0	5.0	pF
			No shield; section 1			Cout	0.37	0.57	pF
			No shield; section 2	6.5	Code E	Cout	0.30	0.46	pF
			No shield			Cga	---	0.90	pF
			No shield			Cgg	---	0.03	pF
		6	No shield			Chk	3.0	5.0	pF
1031	Low-frequency vibration	6	Rp = 2,000 ohms	6.5	Code E	Ep	---	300	mVac
1041	Shock	13	450 G; Ehk = 100 Vdc	6.5	See note 14	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
1031	Vibration-fatigue			6.5	See note 14	---	---	---	---
---	Shock and vibration-fatigue test end points:								
1031	Low-frequency vibration			---	---	Ep	---	450	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	20	uAdc
1256	Electrode current (1) (anode)			---	---	Ib	11.0	---	mAdc
1266	Total grid current			---	---	Ic	0	-0.7	uAdc
1121	Base strain			---	---	---	---	---	---
2126	Envelope strain			2.5	1	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life	4	Ef = 7.5 V; Ehk = 135 Vdc; Ec = Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:			Ihk	---	15	uAdc
1336	Heater-cathode leakage			---	---	---	---
1516	Stability life	6	Eb = Ecc = 100 Vdc; Ehk = 135 Vdc; Rk = 0; Rg/g = 0.5 Meg; TA = room				
---	Stability life-test end point:						
1256	Electrode current (1) (anode)(current in individual tubes)			aIb t	---	10	%
1501	Intermittent life (1) (zero bias)	6,15,16	Eb = Ecc = 100 Vdc; Ehk = 135 Vdc; Rk = 0; Rg/g = 0.5 Meg; TA = room; TE = 140°C (min)	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life (1)-test end points: (zero bias) (500 hours)	17					
1266	Total grid current			Ic	0	-0.5	µAdc
1256	Electrode current (1)(anode) (change in individual tubes)			ΔIb t	---	20	%
1256	Electrode current (1)(anode) (change in individual tubes)			ΔIb Ef	---	15	%
1256	Electrode current (1)(anode) (average change in individual tubes)			Avg ΔIb t	---	15	%
---	Intermittent life (1)-test end points: (zero bias) (1,000 hours)	17					
1266	Total grid current			Ic	0	-0.5	µAdc
1256	Electrode current (1)(anode) (change in individual tubes)			ΔIb t	---	25	%
1256	Electrode current (1)(anode) (change in individual tubes)			ΔIb Ef	---	20	%
1501	Intermittent life (2)	6,15,16	Eb = 250 Vdc; Ec = 0; Ehk = -135 Vdc; Rk/k = 560; Rg/g = 0.5 Meg	---	---	---	---
---	Intermittent life (2)-test end points: (500 hours)	17					
1266	Total grid current			Ic	0	-0.5	µAdc
1301	Heater current			If	420	485	mA
1306	Transconductance			Sm	4,800	8,100	µmhos
1336	Heater-cathode leakage			Ihk	---	7	µAdc
1211	Insulation of electrodes			R	125	---	Meg
---	Intermittent life (2)-test end points: (1,000 hours)	17					
1266	Total grid current			Ic	0	-0.5	µAdc
1301	Heater current			If	420	490	mA
1306	Transconductance			Sm	4,350	8,100	µmhos
1336	Heater-cathode leakage			Ihk	---	7	µAdc
1211	Insulation of electrodes			R	100	---	Meg
1501	Intermittent life (grid pulse)	15,16, 18	Eb = 180 Vdc; Ec = -20 Vdc; egk = 10 V; Rp/a = 200 ohms	---	---	---	---
1501	Intermittent life (grid pulse)-test end point: (500 hours)						
---	Pulse cathode current			Ik	150	---	mA
1511	Cathode interface life	4,16	Ef = 6.9 V	Ri	---	25	Ohms

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NOTES:

1. See "Reduced pressure (altitude) rating," and altitude, maximum peak voltage in the basic document.
2. This value is for operation under fixed-bias conditions. With cathode bias, Rg may be a maximum of 0.5 Meg.
3. The absolute-maximum rating for the combined anode dissipation of both anodes is 4.2 watts.
4. Tie heaters in parallel.
5. For pulse voltage of less than 1 percent duty cycle, the peak voltage shall be a maximum of 150 volts.
6. Test each unit separately.
7. This test shall be peformed at the conclusion of the holding period.
8. Tie each grid to the corresponding anode through a 500,000-ohm resistor (±1 percent).
9. Prior to this test, the tubes shall be preheated 5 minutes with both sections operating at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Ecc	Ebb	Rg/g	Rp/a	Rk/k
V	Vdc	Vdc	Meg	Ohms	Ohms
7.5	180	180	3.9	3,900	0

10. Tie 1k to 2k; 1g to 2g; 1a to 2a.
11. The rejection level shall be set at the VU meter reading obtained during calibration.
12. The positive portion of the grid pulse shall be a square wave meeting pulse shape requirements of method 1296, and, in addition, the maximum amplitude shall occur within the first 20 percent of tp. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect to cathode. Grid resistance not exceeding 50 ohms may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal.

    Peak currents shall be measured by means of a high-impedance oscilloscope, or equivalent device, connected across the cathode resistor. The specified limit refers to the maximum of the pulse amplitude. The variation of the output pulse amplitude between 20 and 80 percent tp shall not exceed the specified limits where tp is defined as in method 1296.
13. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
14. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
15. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurements (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life test-rack.
16. The life-test sample shall consist of 20 tubes and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.
17. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

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NOTES: - Continued.

18. The positive portion of the grid pulse shall be a rectangular wave meeting pulse shape requirements of method 1296. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect to the cathode. Grid resistance not exceeding 50 ohms may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal. The pulse width, tp, shall be 10 ±2 µs, and the duty factor 0.9 to 1.1 percent. Self-excited life-test circuitry is permissible, provided any additional anode voltage drops during the time of pulse are compensated for by increasing Ebb. No fixed Ecl need be applied under self-excited conditions.
19. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 99
DLA - ES

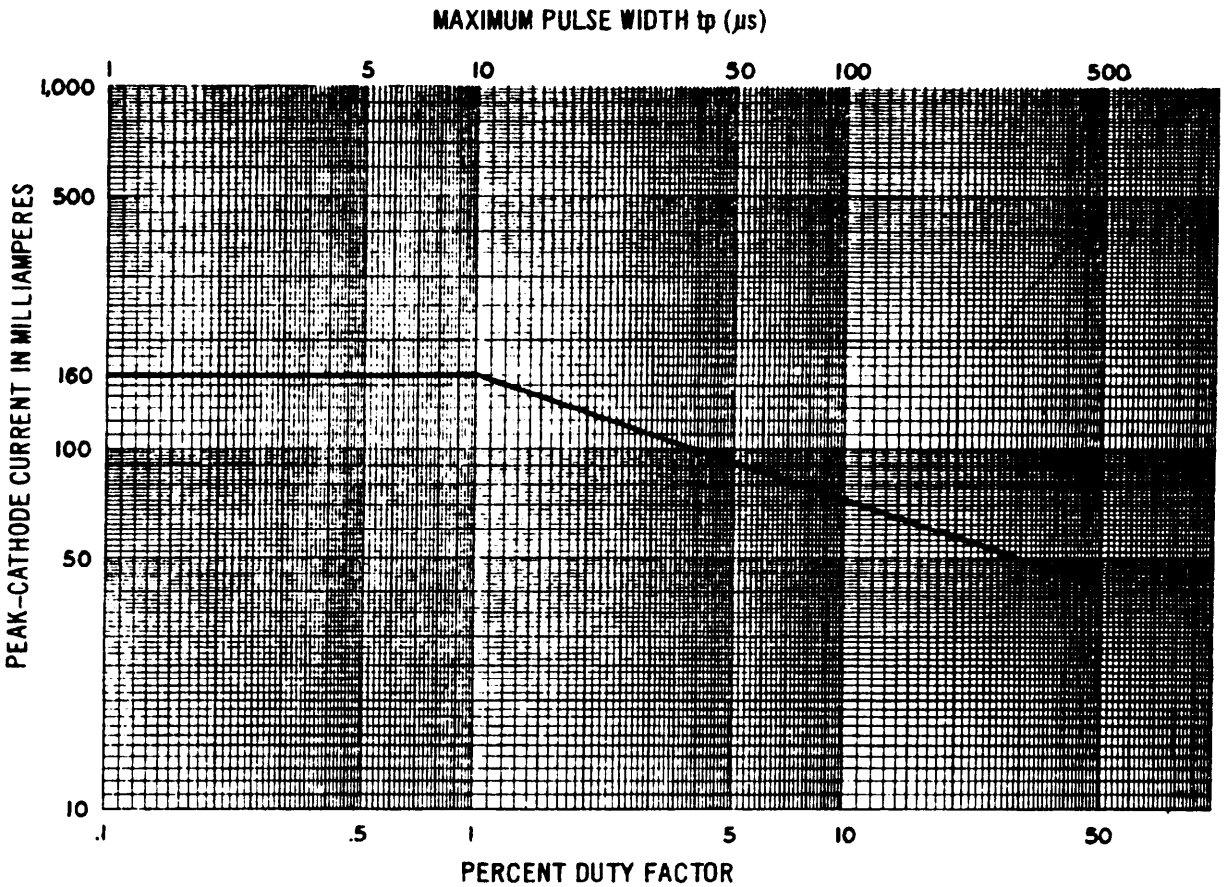
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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The area below and to the left of the line is the area of permissible operation. One hundred and sixty milliamperes is the maximum peak current at any duty factor less than 0.1 percent. For any tube operating under the worst probable conditions, the point indicating peak cathode current and pulse width should both be in the area of permissible operation. Duty factor is defined as the ratio of the average current to the maximum peak current occuring in any 1,000-microsecond period.

FIGURE 1. Pulse rating chart.

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